Same, Same But Encrypted

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Introduction: Wi-Fi "Enhanced Open" is a new security standard for public networks based on Opportunistic Wireless Encryption (OWE). It provides encryption and privacy on open, non-password-protected networks in areas such as cafes, hotels, restaurants, and libraries. WPA3 and Wi-Fi Enhanced Open improve overall Wi-Fi security, providing better privacy and robustness against known attacks. Enhanced Open does not provide authentication.

Contents

1 Lab Prep **1**

2 OWE Configuration and Observation **4**

3 OWE Transition Mode **7**

4 OWE Vulnerabilities **9**

5 General Questions **10**

1 Lab Prep

1.1 Introduction to Opportunistic Wireless Encryption

In January 2018, the Wi-Fi Alliance announced the upcoming release of WPA3. One of the key features disclosed was support for individualized data encryption, which would resolve the long-standing security issues that affect open wireless networks. It was widely speculated that Opportunistic Wireless Encryption (OWE) would be used to implement this feature, and this was later

confirmed when WPA3 certification began in June 2018.

By early 2019, WPA3's implementation of OWE had been rebranded as "Wi-Fi Certified Enhanced Open" and converted to a separate and optional standard to be released alongside WPA3. Yet in spite of its status as an optional requirement for hardware certification, OWE was still a major source of excitement among wireless professionals and enthusiasts alike. Multiple major device manufacturers announced current or future support for OWE by the Spring of 2019.

OWE establishes a secure connection over an unsecure medium using the Diffie-Hellman (DH) key exchange in combination with an Elliptic Curve Group (mandatory for OWE) or a Prime Modulus Group (non-mandatory). The Association Frames now include public keys (i.e., DH parameters), one from the client and one from the AP. As encryption requires keys, the main difference to the open connection establishment is the added EAP-key handshake to derive a secret session key for data encryption. After the EAP key message No. 4, encrypted data frames are exchanged between an AP and a client. The overall OWE authentication, association and key exchange procedure is illustrated in Fig. 1.

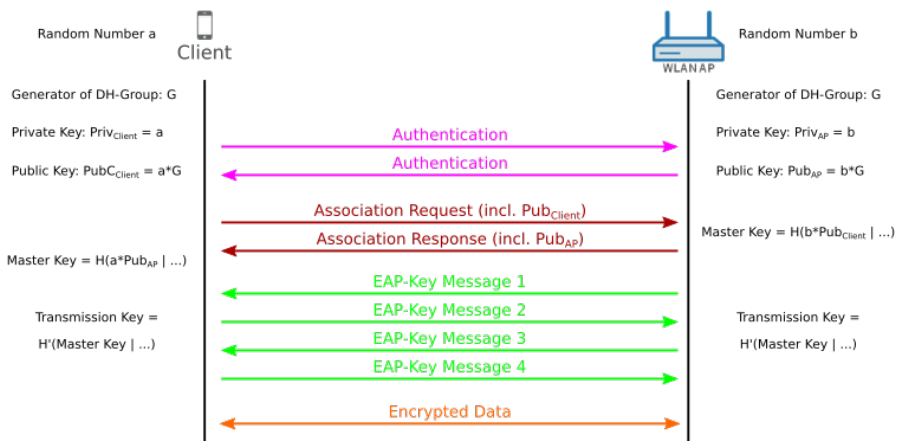


Figure 1: OWE flowchart.

1.2 Protected Management Frames

All Enhanced Open devices need to use Protected Management Frames (PMF), so it is activated implicitly as soon as the user connects to an Enhanced Open SSID. PMF delivers a protection against forging management frames, e.g. an attacker can disassociate any user by claiming to be the Access Point (AP) that the client is currently connected to. Enhanced Open uses the "Management Frame Protection Required" mode with SHA-256 for Hashes. Although it has been around for a longer time, the market adoption was rather low.

1.3 Install wpad-openssl

In the following subsections, we will install WPA3 related packages on a router. Since the default wpad-basic/wpad-mini package does not support WPA3. We will need to remove wpad-mini first and then install the full version of wpad-openssl. **Your router needs to have Internet to install the software package.**

Log onto the router as the root user via ssh, and issue the following commands:

```
opkg update

opkg remove wpad-basic
# Or opkg remove wpad-mini (if appropriate)
```

An exmmple screenshot is shown in Fig. 2.

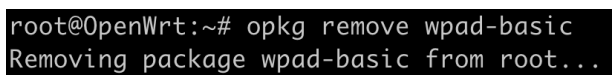


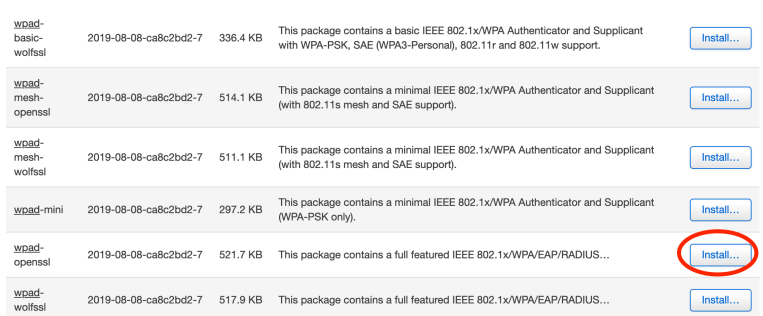
Figure 2: Remove wpad-basic.

Next, install wpad-openssl as follows.

```
opkg install wpad-openssl
```

It should be okay the opkg update command gives some warnings message. However, the installation of wpad-openssl needs to be successful. Reboot your router to make the newly installed software effective.

If you have difficulties using the command line to install wpad-openssl, you can use the web interface to install it. Go to System -> Software and search for wpad after updating the list first. Then, select wpad-openssl to install (an example is shown in Fig. 3).



wpad-basic-wolfssl	2019-08-08-ca8c2bd2-7	336.4 KB	This package contains a basic IEEE 802.1x/WPA Authenticator and Supplicant with WPA-PSK, SAE (WPA3-Personal), 802.11r and 802.11w support.	Install...
wpad-mesh-openssl	2019-08-08-ca8c2bd2-7	514.1 KB	This package contains a minimal IEEE 802.1x/WPA Authenticator and Supplicant (with 802.11s mesh and SAE support).	Install...
wpad-mesh-wolfssl	2019-08-08-ca8c2bd2-7	511.1 KB	This package contains a minimal IEEE 802.1x/WPA Authenticator and Supplicant (with 802.11s mesh and SAE support).	Install...
wpad-mini	2019-08-08-ca8c2bd2-7	297.2 KB	This package contains a minimal IEEE 802.1x/WPA Authenticator and Supplicant (WPA-PSK only).	Install...
wpad-openssl	2019-08-08-ca8c2bd2-7	521.7 KB	This package contains a full featured IEEE 802.1x/WPA/EAP/RADIUS...	Install...
wpad-wolfssl	2019-08-08-ca8c2bd2-7	517.9 KB	This package contains a full featured IEEE 802.1x/WPA/EAP/RADIUS...	Install...

Figure 3: Install wpad-openssl via web GUI.

2 OWE Configuration and Observation

2.1 Configure an OWE Network

Next, we proceed to configure a WPA3 OWE enforced wireless network. Select an SSID (i.e., the network name) and a secret for your wireless network. Issue a set of uci commands according the following prototypes (fill in the values based on your actual situation) to enable OWE.

```
uci set wireless.default_radio1.mode='AP'
uci set wireless.default_radio1.ssid='SSID-of-your-choice'
uci set wireless.default_radio1.encryption='owe'
uci commit wireless && wifi
```

You can verify the configurations via the uci command with the show option (i.e., uci show wireless). Note that there is no secret supplied in OWE.

If you have difficulties using the command line to configure WPA3 with OWE, you can use the web interface to configure it. Go to Network -> Wireless, and select an wireless interface to configure. Under Wireless Security, choose OWE (an example is shown in Fig. 4). In addition, the Management Frame Protection feature is enabled by default.

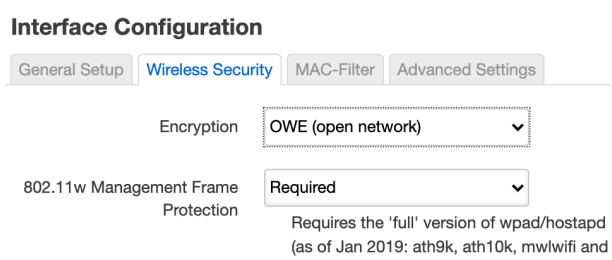


Figure 4: Enable WPA3 OWE via web GUI.

To verify that your OWE network is properly configured, capture a beacon frame of your wireless network using Wireshark. Open the beacon frame, and expand the "Tagged parameters" menu. Next, expand the "RSN Information" tag. And then expand the "Auth Key Management List -> Auth Key Management Suite (AKM) Suite" tags. Notice the value of the "Auth Key Management (AKM) type" element shown in Fig. 5. APs that support OWE advertise their support for the protocol using an Authentication and Key Management (AKM) Suite Selector that is added to the RSN element of all beacons and probe response frames that the AP issues.

2.2 OWE Exchange Observation

Now that we have a WPA3 OWE enabled wireless network. We will have a device connect the network and observe the key steps in OWE. Have a WPA3 OWE ready device connect to your wireless network (note that not all devices support OWE yet). Note that there is no secret needed

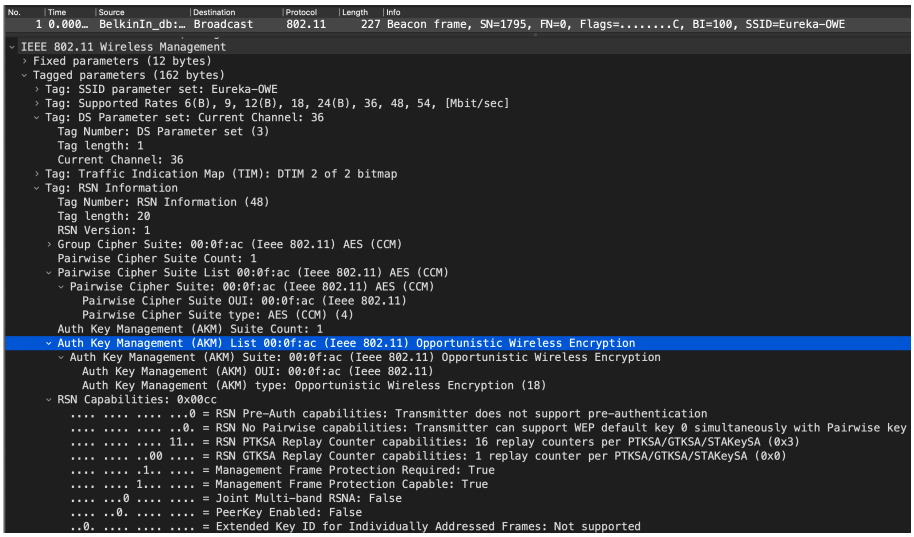


Figure 5: A beacon frame of the OWE network.

when connecting to the network. At the same time, start capturing wireless frames (via Wireshark, Airodump-ng or Airtool) on the right channel to capture the OWE authentication and key derivation frames.

First, we will observe the overall OWE exchange process. Open the frame capture file in Wireshark, and locate the series of OWE exchange frames. The sequence of frames starts with an open authentication message and ends with a session key generation frame as shown in Fig. 6. A complete example of captured frames is available at `http://csfaculty.tcu.edu/ma/eureka/?dir=WPA3-OWE`.

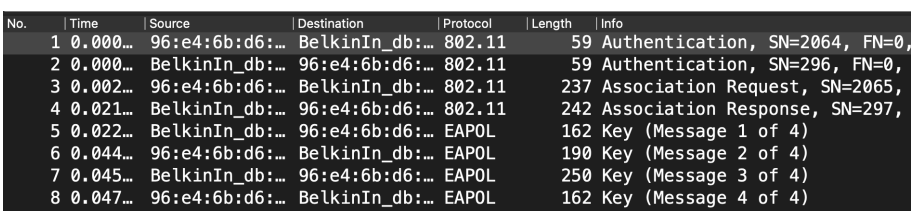


Figure 6: Complete OWE exchange sequence captured (matching the flowchart).

2.2.1 Provide a screenshot of your captured OWE exchange sequence.

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Authentication (Open System): This is the same as the one used in connection establishment of an open network. An example of the first authentication frame is shown in Fig. 7 (open system authentication is a just a fancy way of saying "no authentication actually occurs"). You can see the authentication process taking place in Wireshark using the following filter: wlan.fc.type_subtype == 0xb.

2.2.2 Provide screenshots of your captured authentication frames for OWE.

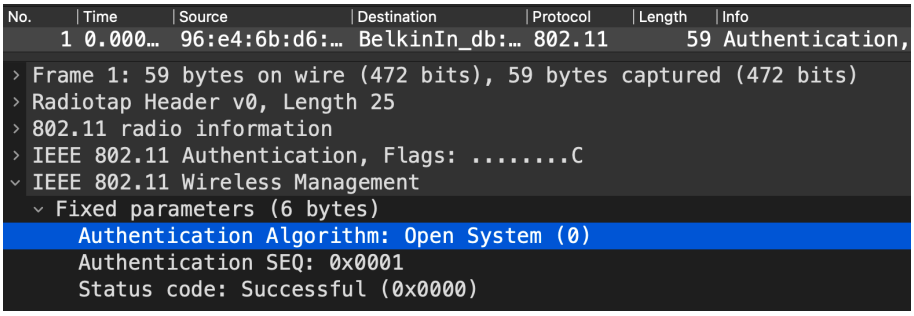


Figure 7: Authentication frame.

OWE DH Public Parameter Exchange: Once a client has performed open system authentication with the AP, the client and AP transition to the 802.11 association process. It is during association that OWE's Diffie-Hellman (DH) Key Exchange and 4-way handshake occur. As indicated in Fig. 8, the Association Request and Response contain additional information – the public key material of client and AP (an extended tag named "OWE Diffie-Hellman Parameter"). Before the Association takes place, both sides choose a random number (its private key from now on) and multiply it by the Generator G of the chosen DH-Group to retrieve the public key.

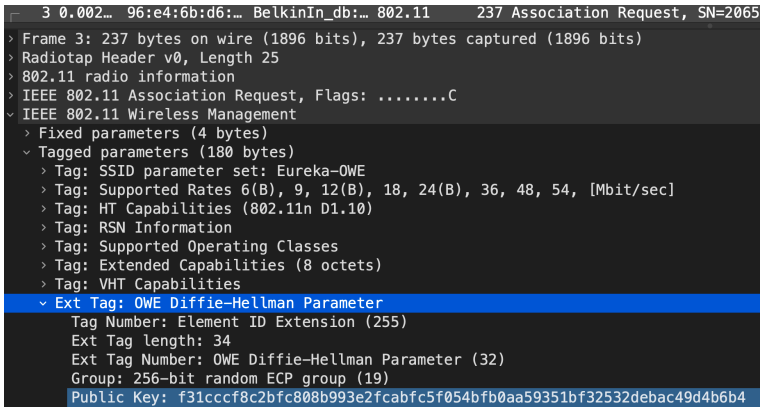
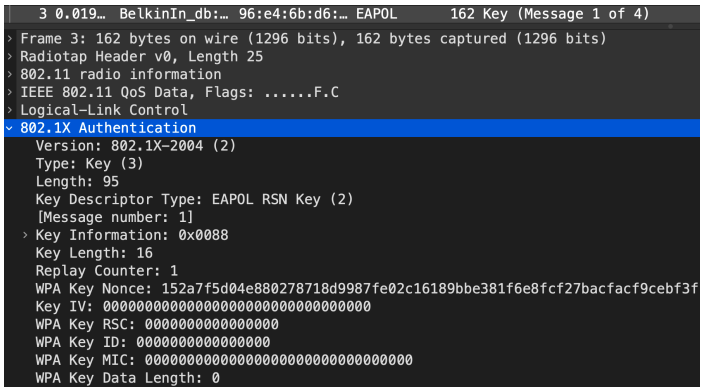


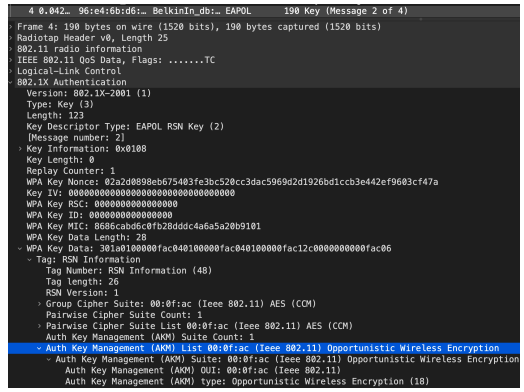
Figure 8: Exchange of DH public key parameters in Association Request.

2.2.3 Provide screenshots of your captured association request and response frames for OWE.

OWE Session Key Derivation: Once the client and AP have completed the association process as well as the DH Key Exchange, the Pairwise Master Key (PMK) is created. The PMK is assigned a unique identifier known as a Pairwise Master Key Identifier (PMKID). The AP then initiates a 4-way handshake with the client in which the PMK is used to create encryption keys. There will



(a) Key agreement exchange message 1 of 4.



(b) Key agreement exchange message 2 of 4.

Figure 9: OWE session key derivation frames.

be four frames in the key derivation procedure. The first two frames are shown in Fig. 9.

2.2.4 Provide screenshots of your captured frames for OWE key agreement/derivation frames.

2.2.5 Launch a dis-association attack using Aircrak-ng. Verify whether the protected frame feature works.

3 OWE Transition Mode

It is expected that "legacy" clients that do not support OWE will exist for a long period of time. A transition mode is designed to accommodate legacy clients, which allows to configure an open SSID with an Enhanced Open "Buddy"-SSID. With the introduction of the transition mode, network operators can easily offer this new security standard right away, without the fear of loosing any legacy clients.

To enable the transition mode, you will need to use two wireless interlaces on the router. One interface is host an OWE enabled network, and the other is to host an open network. A client will connect to the open network first, and transition to the OWE network.

In the following example, assume the first interface (commonly shown as "radio0") on your router will be used for the open network (whose SSID is Eureka-Open), and the second interface (commonly shown as "radio1") on your router will be used for the OWE network (whose SSID is Eureka-WPA3-OWE).

Next, we proceed to configure an open wireless network on the first wireless interface (i.e., radio0/wlan0). Issue a set of uci commands according the following prototypes (fill in the values

based on your actual situation).

```
uci set wireless.default_radio0.mode='AP'
uci set wireless.default_radio0.ssid='Eureka-Open'
uci set wireless.default_radio0.encryption='none'
uci set wireless.default_radio0.owe_transition_ssid='Eureka-WPA3-OWE'
uci set wireless.default_radio0.owe_transition_bssid='MAC-Addr-of-Radio1'
uci commit wireless && wifi
```

Note that there are two additional options: owe_transition_ssid and owe_transition_bassid, which refer to the OWE network's SSID and the MAC address (BSSID) of its hosting wireless interface. You can verify the configurations via the uci command with the show option (i.e., uci show wireless).

Now, configure OWE on the second wireless interface (i.e., radio1/wlan1). Issue a set of uci commands according the following prototypes (fill in the values based on your actual situation) to enable OWE. The set of commands are identical to the ones used in Section 2.1. There is an additional option that hides the OWE network.

```
uci set wireless.default_radio1.mode='AP'
uci set wireless.default_radio1.ssid='Eureka-WPA3-OWE'
uci set wireless.default_radio1.encryption='owe'
uci set wireless.default_radio1.hidden='1'
uci commit wireless && wifi
```

You can verify these settings by checking from the web GUI. An example is shown in Fig. 10.

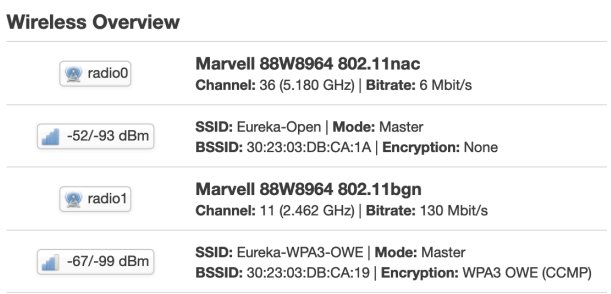


Figure 10: Enable WPA3 OWE transition mode via web GUI.

If you have difficulties using the uci command to configure OWE, you can use vi to edit the "/etc/config/wireless" file on the router directly.

Once the configuration is completed, have an "Enhanced Open capable" client connect to the open network. Also, start a capture (via Wireshark, Airodump-ng or Airtool) to collect all the frame exchanges.

The beacon frame of the open network, as shown in Fig. 11 with the SSID of “Eureka-Open” in our example, contains new information. A vendor specific information element with the title “Wi-Fi Alliance: OWE Transition Mode” is added to it, which tells a client about the OWE “Buddy”-SSID called “Eureka-WPA3-OWE” and its BSSID. An Enhanced Open capable client will use this information from the Beacon and start a (unicast) Probe Request directed to the hidden SSID (BSSID) to connect to the encrypted open network automatically. Client devices that do not support OWE will ignore the OWE Transition mode element and simply connect to the open network like before.

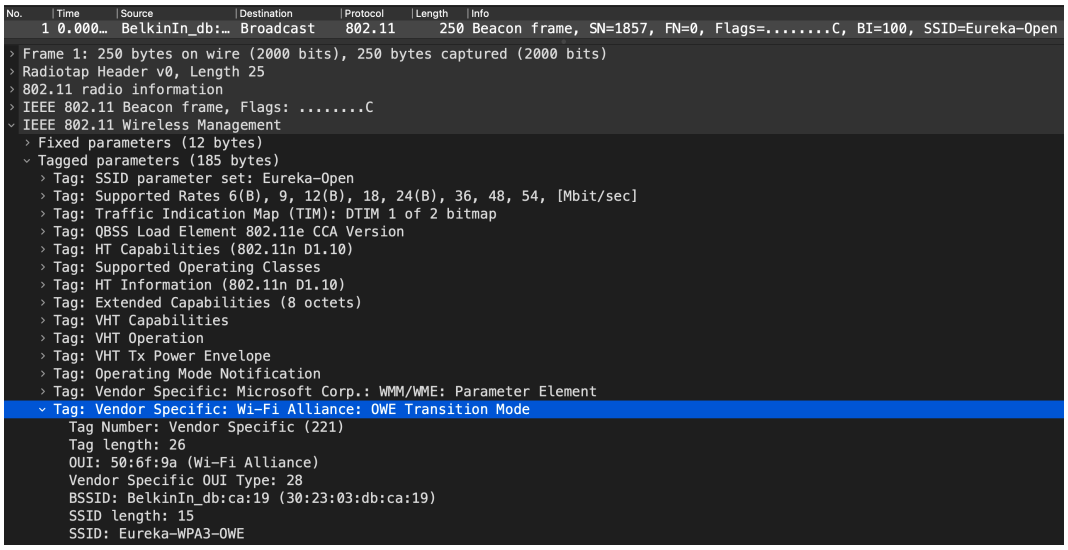


Figure 11: A beacon frame of the OWE transition mode.

3.0.1 Capture a beacon frame of your OWE transition network showing a vendor specific element of type “OWE Transition Mode”. Provide a screenshot of the captured frame with the required information.

3.0.2 Observe the transition process. Capture a Probe Request directed to the hidden SSID.

4 OWE Vulnerabilities

The introduction of OWE by the WiFi Alliance is a step in the right direction, in that it attempts to fill a need that is now two decades old — encryption support for unauthenticated wireless networks.

However, OWE addresses sniffing attacks, but not rogue AP attacks. There are two major reasons: i) OWE does not provide a mechanism for validating the identity of an AP; ii) RFC compliant supplicants actively prevent users from distinguishing between encrypted and unencrypted connections.

4.0.1 Recall the KRACK attack that breaks WPA2 without obtaining the real key (Passphrase). Enhanced Open uses the same EAPoL 4-Way Handshake like WPA2 (and WPA3) to exchange transmission keys. Is Enhanced Open also vulnerable to the KRACK attack? Why or why not?

4.0.2 Is OWE Transition Mode susceptible to abuse and attack, and if so, how?

4.0.3 (Optional) Provide step-by-step walk-through of a proof-of-concept attack against OWE. You need to provide sufficient technical details.

5 General Questions

5.0.1 What is the most challenging part in this lab?

5.0.2 Which part of this lab helps you better understand the concepts covered in the lecture?

5.0.3 Are the lab activities fun and inspiring learning?

5.0.4 Are there any corrections/improvements needed to apply to this lab?